

M52: Final Composite Dependent Variable

PCA on Raw Switch Counts (Time, Topic-LDA, Typing)

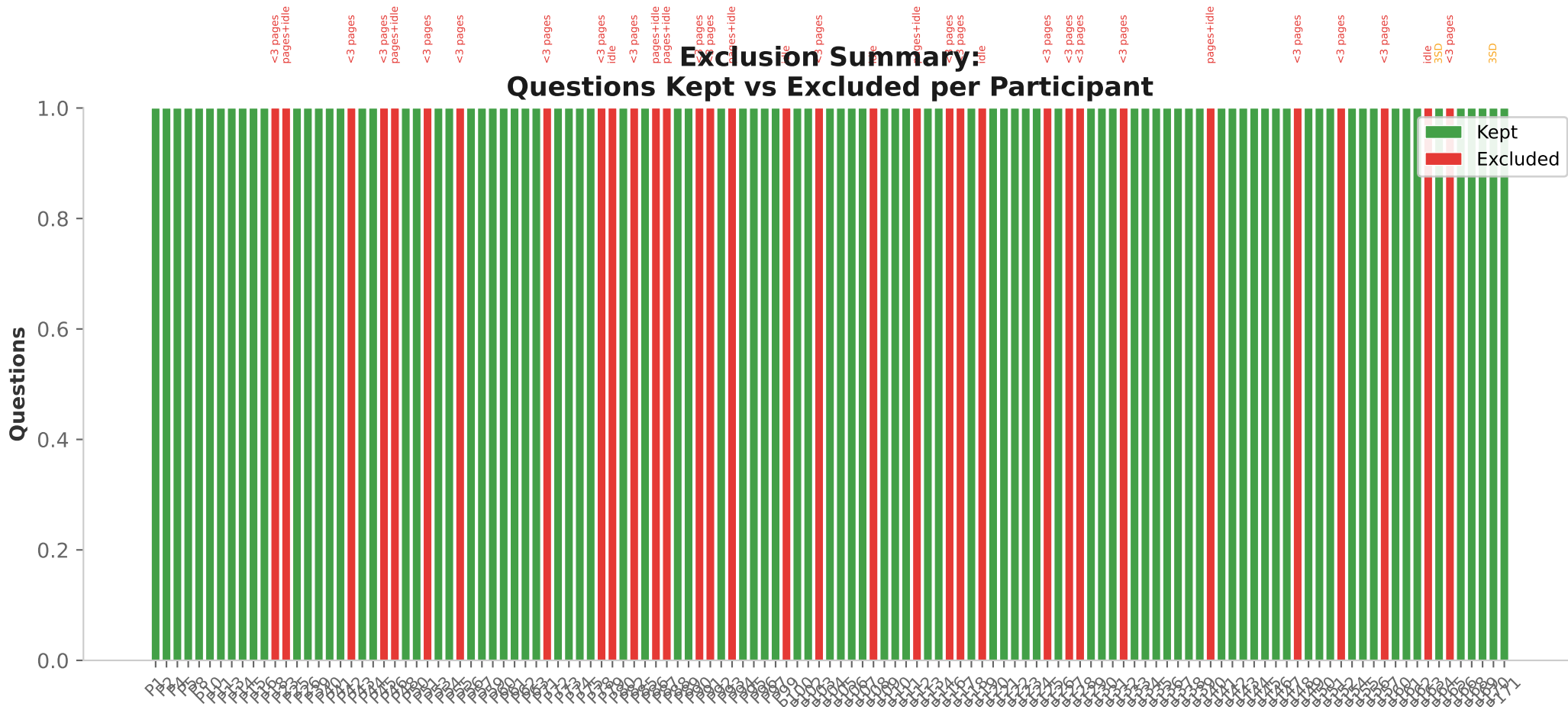
No standardization (z-score) applied

Exclusion Criteria Applied:

1. Questions with fewer than 3 page visits: 28 excluded
2. Questions with idle time $\geq 50\%$ after last meaningful event: 12 excluded
3. Participants > 3 SD from mean on any DV: 2 excluded

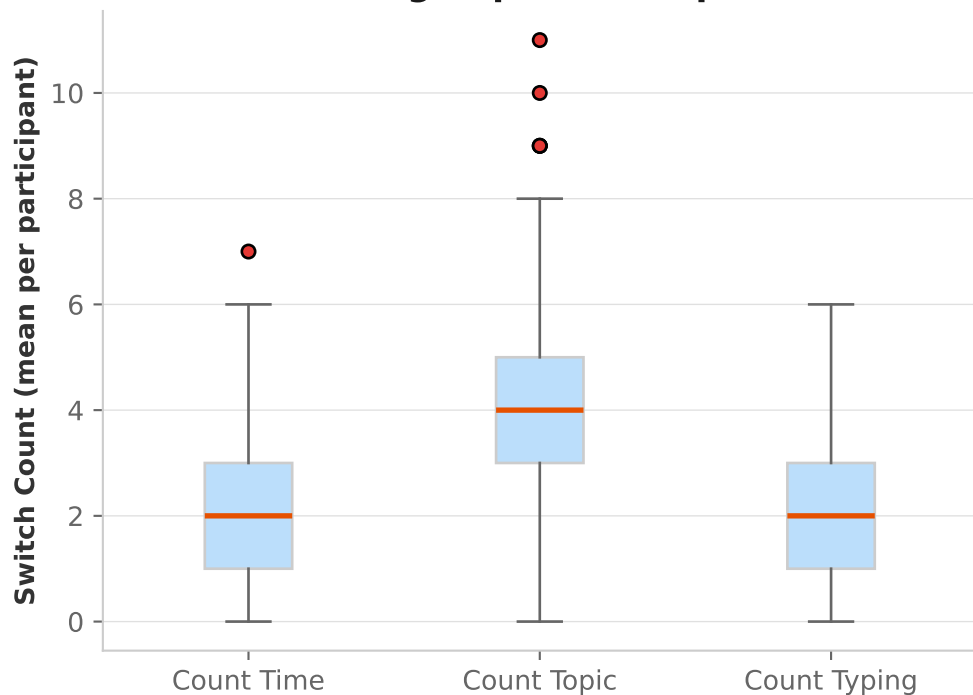
Result: 92 / 125 questions retained

Final sample: N = 90 participants

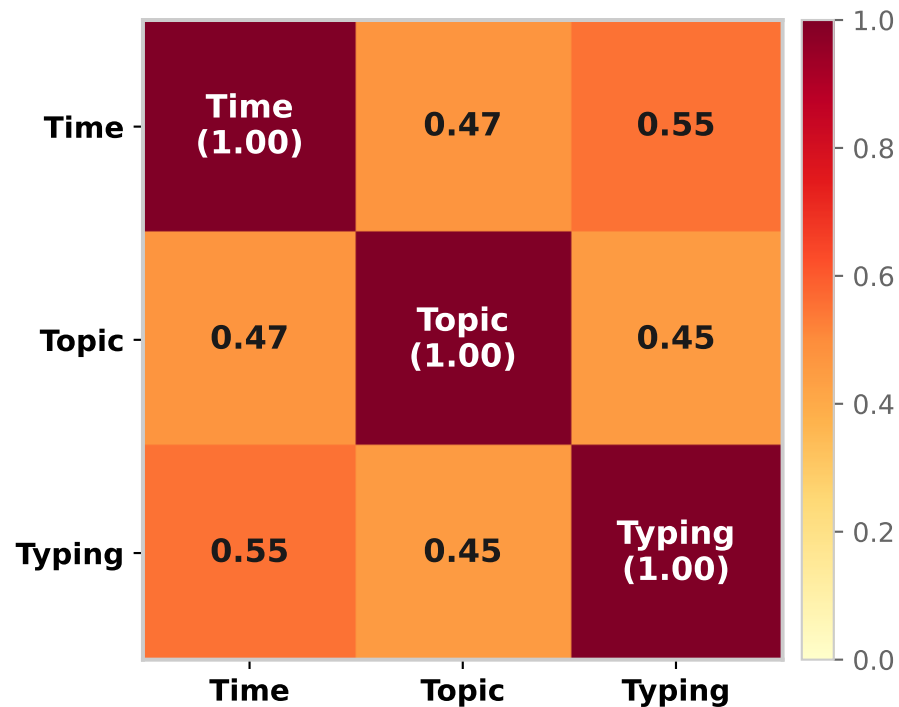


Descriptive Statistics: Raw Switch Counts

Distribution of Raw Switch Counts (Averaged per Participant)

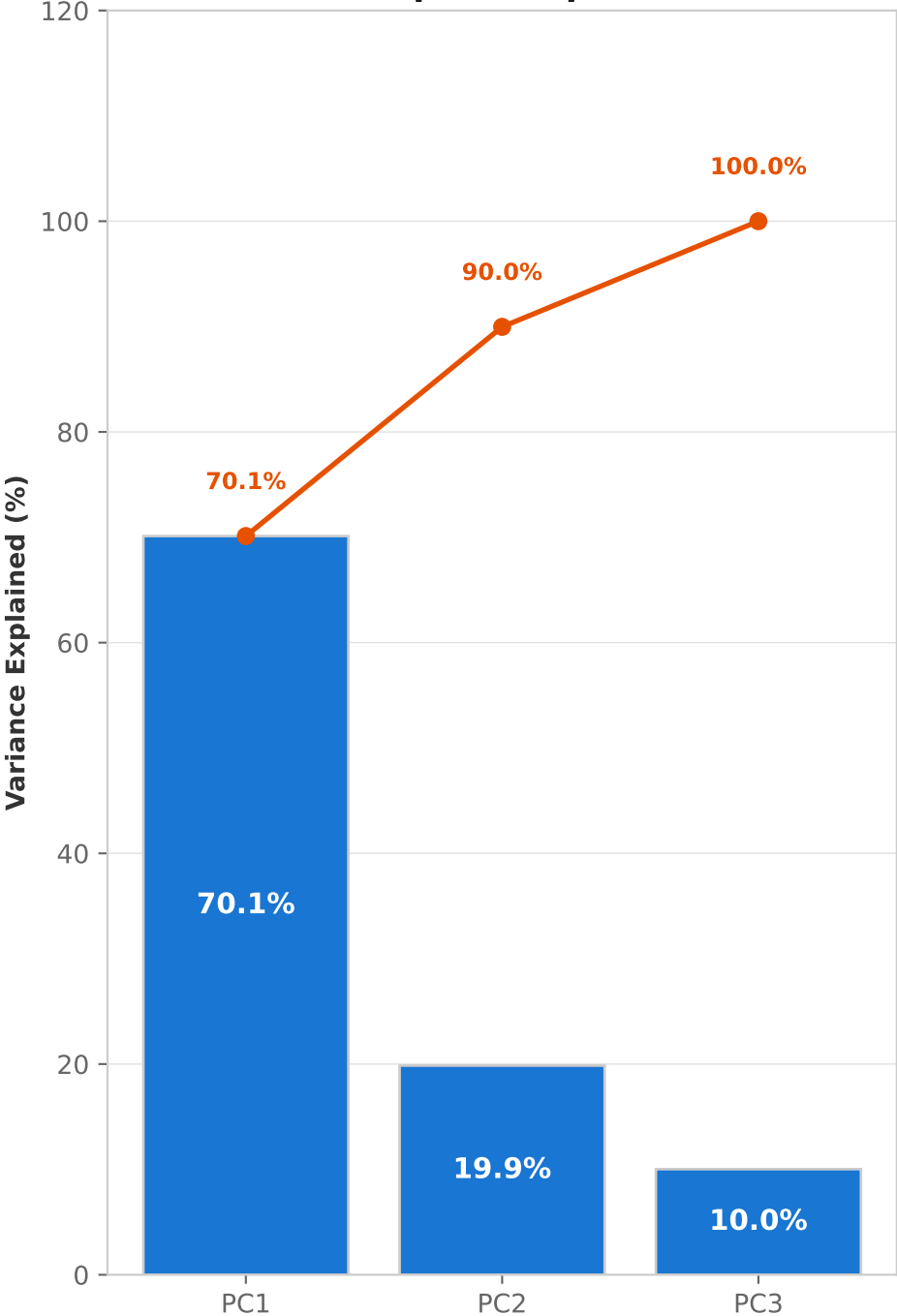


Correlation Between Switch Count Variables



PCA Results: Dimensionality Reduction on Raw Switch Counts

Variance Explained by Each Principal Component



PCA Loadings (variable weights per component):

PC1 (70.1% variance):

Count Topic	+0.839	+++++
Count Time	+0.401	+++++
Count Typing	+0.368	+++++

PC2 (19.9% variance):

Count Time	+0.611	+++++
Count Typing	+0.575	+++++
Count Topic	-0.545	-----

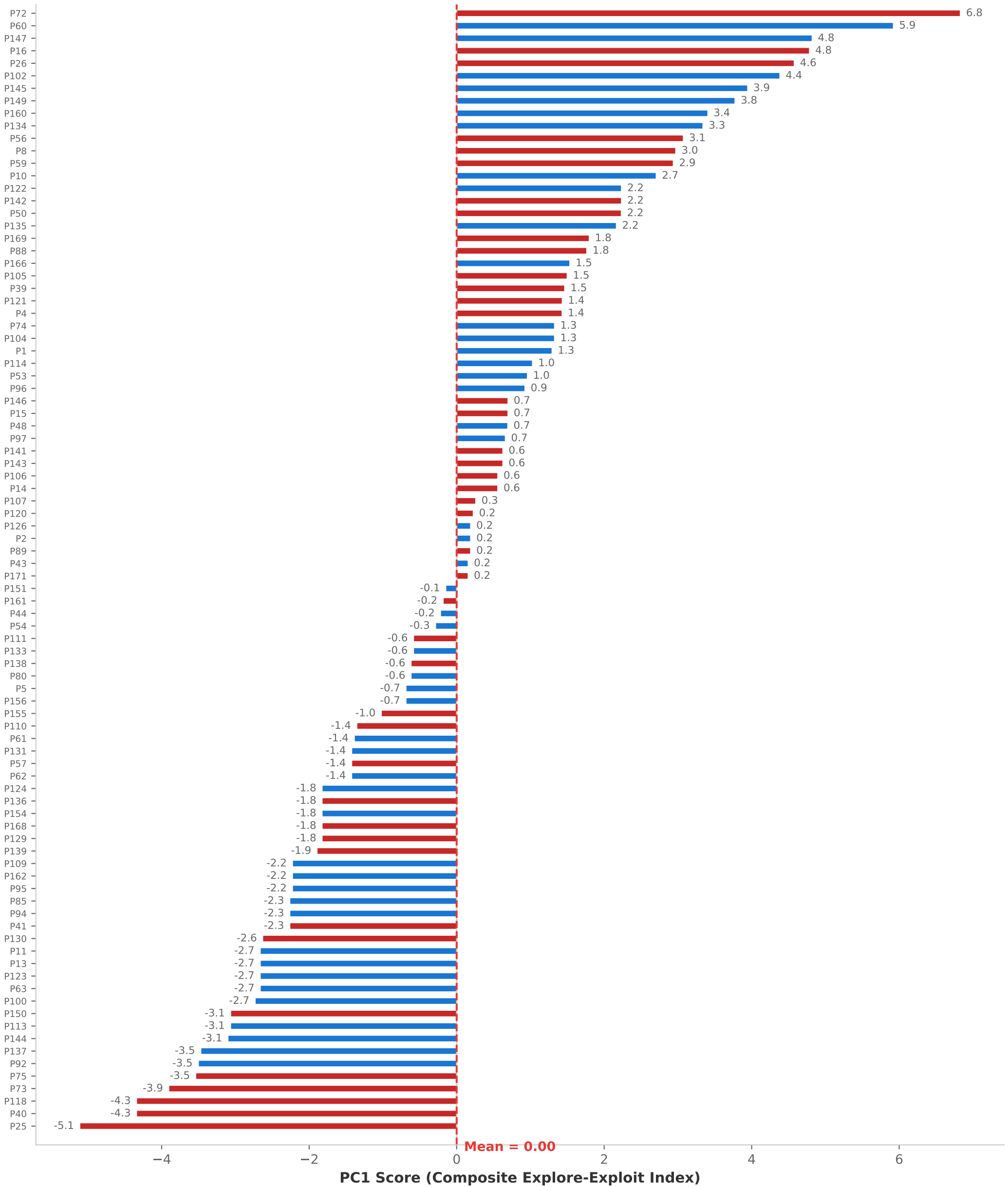
PC3 (10.0% variance):

Count Typing	+0.731	+++++
Count Time	-0.682	-----
Count Topic	+0.006	

Interpretation:

- PC1: Primarily reflects topic switching (loading = +0.839)
Higher score = more topic transitions
- PC2: Primarily reflects time switching (loading = +0.611)
Higher score = more time transitions
- PC3: Primarily reflects typing switching (loading = +0.731)
Higher score = more typing transitions

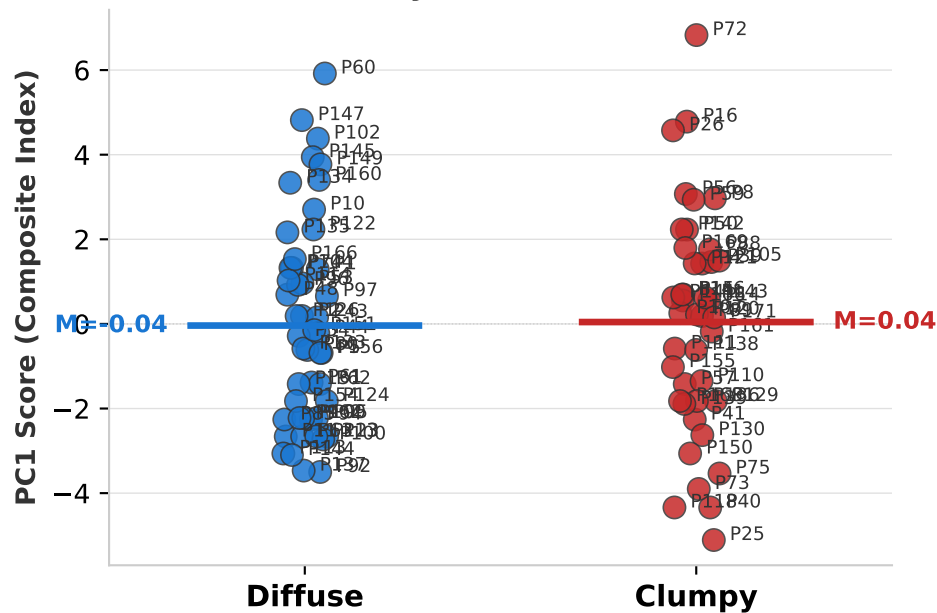
Individual PC1 Scores: Composite Explore-Exploit Index



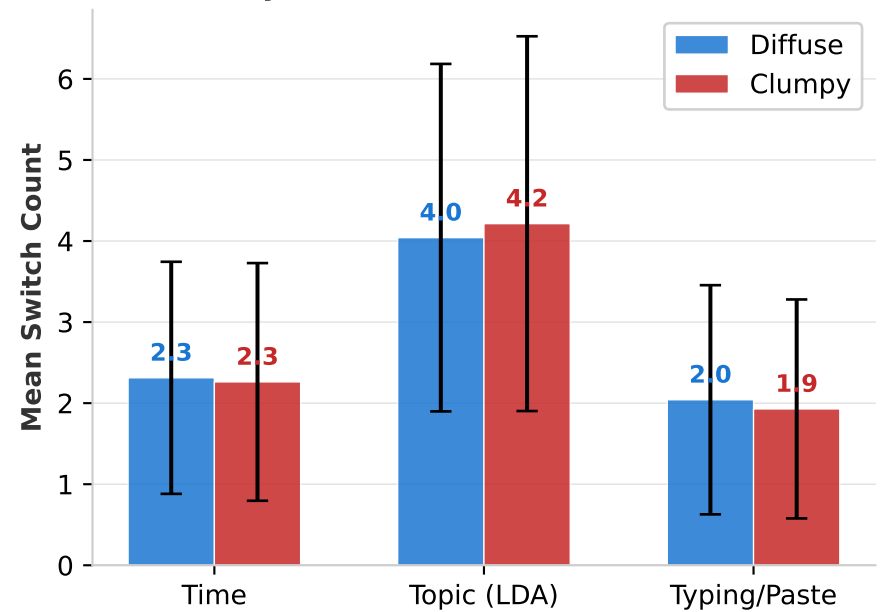
Condition Comparison: Clumpy vs Diffuse

N = 90 (Diffuse: 48, Clumpy: 42)

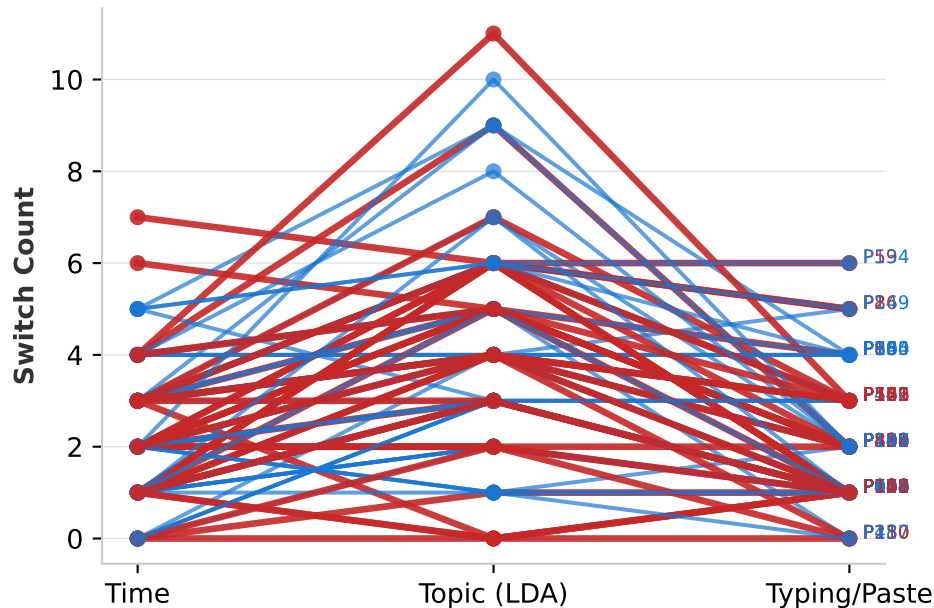
PC1: Composite Explore-Exploit
by Condition



Raw Switch Counts
by Condition (Mean +/- SD)



Individual Profiles
by Switch Type



Conditions

- Diffuse (n=48)
- Clumpy (n=42)

Condition Comparison: Descriptive Statistics & t-Tests

N = 90 (Diffuse: 48, Clumpy: 42)

DESCRIPTIVE STATISTICS

Variable	Diffuse (n=48)	Clumpy (n=42)
Time	M = 2.31, SD = 1.43	M = 2.26, SD = 1.47
Topic (LDA)	M = 4.04, SD = 2.14	M = 4.21, SD = 2.31
Typing/Paste	M = 2.04, SD = 1.41	M = 1.93, SD = 1.35
PC1 (composite)	M = -0.04, SD = 2.43	M = 0.04, SD = 2.60

INDEPENDENT-SAMPLES t-TESTS (Welch)

Variable	t	df	p	Cohen d	Sig.
Time	0.17	85.8	0.8692	-0.03	n.s.
Topic (LDA)	-0.37	84.3	0.7156	+0.08	n.s.
Typing/Paste	0.39	87.3	0.6991	-0.08	n.s.
PC1 (composite)	-0.16	84.5	0.8767	+0.03	n.s.

* p < .05 ** p < .01 *** p < .001

INTERPRETATION

PC1 difference (Clumpy - Diffuse): +0.08
Clumpy condition shows more explore-exploit switches
Effect size (Cohen's d): +0.03
Result is NOT statistically significant (p = 0.8767)
Low statistical power due to small sample (N = 90)

Largest sub-variable effect: Typing/Paste
d = -0.08, p = 0.6991

Note: Typing metric uses answer_snapshot bursts + paste events.